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Satellite tracking of Humboldt penguins (*Spheniscus humboldti*) in northern Chile

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Abstract During the El Niño of 1982/1983, the Humboldt penguin population diminished dramatically in the whole distributional area of the species. Recovery of the population was slow since 1983 and it has been suggested that large numbers of Humboldt penguins die at sea, entangled in nets, or starve to death, even during non-"El Niño" years. We were able to determine for the first time, how Humboldt penguins on Pan de Azúcar Island (26°S; 72°W) utilize their marine habitat and where their feeding areas lie. For this purpose we employed two streamlined Argos satellite transmitters during the 1994/1995 and 1995/1996 breeding seasons, respectively. Mean travelling speed of Humboldt penguins during foraging trips was 0.94 m s^{-1} and 50% of bird positions were located within 5 km of the island (90% within 35 km). Total area covered by Humboldt penguins foraging from Pan de Azúcar Island was 12 255 km^2. Satellite transmitters also recorded dive duration; penguins spent on average 7.8 to 9 h diving per foraging day but showed no preferences for particular feeding areas. Mean daily dive durations (4-d mean) recorded during the 1994/1995 breeding season were positively correlated between birds. Significant correlation between dive duration and sea surface temperature anomalies and negative correlation between dive duration and fishery landings at nearby Caldera harbour indicate that the 1994/1995 increase in foraging effort was a response to deteriorating prey availability. Sea surface temperatures during the 1995/1996 breeding season were colder than average, and we observed no trends in bird diving activities.

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Introduction

The Humboldt penguin is distributed along the Peruvian and Chilean coasts, from 5°S to about 42°S (Williams 1995). It exploits the high marine productivity in the area, which itself is based on the nutrient-rich Humboldt Current flowing northward along the Pacific coast. Humboldt penguin numbers are severely affected by El Niño (EN) events. Hays (1986) reported 65% mortality of the initial population following the 1982/1983 EN. Latest estimates range on the order of 10 000 animals surviving to date (Araya personal communication). Although the species has been included on the "red list" in Chile as well as in Peru, very little is known about its biology, its energetic demands and its behaviour, either on land or at sea (Williams 1995).

Competition with commercial fisheries has been suggested (Wilson et al. 1988) as the main cause of the continuing decrease of the related African penguin (*Spheniscus demersus*), which is the only *Spheniscus* species for which comprehensive at-sea distribution data exists (e.g. Wilson and Wilson 1990). In order to determine the biotic and abiotic characteristics of the areas in which Humboldt penguins forage, and to establish commercial fishing limits around the few islands that are still used by breeding colonies, we wanted to investigate where Humboldt penguins from one of the main breeding sites, Pan de Azúcar Island (located in Pan de Azúcar National Park), Chile, search for prey.

While transects conducted at sea to determine the range of penguins are costly, time-consuming and labour-intensive, telemetric devices offer an opportunity to track individual birds. Because the operational range of radio transmitters is limited, and the operation of the receiving antennas are labour- and time-intensive, we used four miniaturized, streamlined satellite transmitters. These were equipped with a salt water switch, which served two purposes: to reduce power consumption of the devices, by turning the devices off while the birds were submerged, and to provide data for a counter of submersion time, which was read out every time data transmission was successful.

Materials and methods

Field work was conducted in northern Chile at Pan de Azúcar Island in Pan de Azúcar National Park (26°09′S; 70°40′W) between November 1994 and January 1995 and between November 1995 and January 1996. During both breeding seasons, two penguins each were caught by hand at their nest site after administration of 0.25 ml ketamine hydrochloride (Ketavet, Parke-Davis GmbH, Berlin, Germany) to tranquilize the birds [assuming a bird mass of approximately 5 kg, and employing the dose of 5 mg kg^{-1} suggested by Luna-Jorquera et al. (1997a)]. The birds were then equipped with ST 10 PTT (platform transmitter terminal) satellite transmitters (Telonics, Mesa, Arizona, USA) weighing 130 g with dimensions $140 \times 48 \times 20$ mm. Each transmitter was equipped with a 220×2 mm antenna attached to the back at an angle of 45° with the body of the unit. The ST 10 PTTs were streamlined by attaching a wedge-shaped "nose" (polyurethane foam, pressure resistant to 50 bar) to the front end, shaped according to the suggestions of Bannasch et al. (1994). The streamlined units were attached to the back of the birds in a position as caudal as possible without impairment of the tail or the preen gland (cf. Culik et al. 1994), by using tape (Tesa-Band, Beiersdorf, Germany) secured to the feathers according to the methods described by Wilson and Wilson (1989) and Wilson et al. (1997), and a water-resistant rubber cement (Pattex, Henkel, Düsseldorf, Germany).

All satellite transmitters used in this study were equipped with a salt water switch, preventing transmission and thus saving power while the penguins were under water. The salt water switch was checked every 1/250 s, and transmission was initiated within a few milliseconds after surfacing of the individual. Thereafter, transmission interval was 60 s, as long as the penquin remained surfaced. The salt water switch was connected to a timer, which registered cumulative dive durations for up to 18.2 h before tripping over and starting again from 0. The actual data of the timer was transmitted whenever the birds were at the surface. Because we received several messages per day from the transmitters, this was sufficient to deduce daily dive duration.

Performance of the satellite transmitters

Transmitters used during the 1994/1995 season were of two types. While bird no. 1 carried a transmitter which, while at the surface, operated continuously, penguin no. 2 was equipped with a transmitter which worked on a 16 h on, 8 h off schedule. Positional information was received on average 4.16 times per day (± 1.35 SD, $n = 38$ d) from transmitter no. 1, while transmitter no. 2 (started at 1500 hrs GMT or 1100 hrs local time) was only received 3.05 times daily (± 1.08 SD, $n = 55$ d). However, the 16 h on, 8 h off schedule increased the life span of transmitter no. 2 by 45% (55 d vs 38 d) over that of transmitter no. 1.

Transmitters used during the 1995/1996 season (started at 0800 hrs GMT or 0400 hrs local time) were both programmed to transmit on a 16 h on, 8 h off basis. However, we only received 1.78 (± 0.89 SD, $n = 45$ days) and 1.68 (± 0.97, $n = 63$ d) locations per day from bird 17 and 115, respectively.

Location data received from Argos (CLS, Toulouse, France) is ranked according to quality. Location accuracy for class a and b locations is not provided, and the user is requested to assess the quality of these data based on previous locations of the PTT, assumed speed of movement and elapsed time since the last location of class 0 to 3. Location classes obtained from the four transmitters used in this study (512 locations in total) are summarized in Table 1. There was no difference between transmitter types (i.e. programmed versus free-running) or years. On average, a total of 38.8% of all locations were obtained with an accuracy of 1000 m or less (i.e. class 1 or better).

Table 1 Performance of Argos satellite transmitters: frequency of Argos locations obtained from the most accurate class to the least accurate class, and classes a and b, for which location accuracy is evaluated by the user

Class	Frequency (%)	Accuracy (m)
3	3.1	150
2	11.1	150–350
1	24.6	350–1000
0	27.1	> 1000
a	16.8	
b	17.2	

Results

Birds and device effects

Details on the Humboldt penguins used in this study are given in Table 2. Both birds used in 1994 were equipped on 10 November. The nest of bird no. 2 was abandoned on 22 November, while that of bird no. 1 was abandoned on 26 November. The transmitter on penguin no. 1 worked until 19 December while data from transmitter no. 2 were received until 3 January 1995.

Both Humboldt penguins studied in the 1995/1996 season (nos. 17 and 115) were equipped on 15 November. Both adults carrying PTTs were also used for a study of field metabolic rate using doubly labelled water, and blood samples were taken each time the birds were

Table 2 *Spheniscus humboldti*. Penguins equipped with satellite transmitters at Pan de Azúcar Island, northern Chile. Data on chick mass refer to chick 1/chick 2. Birds equipped with transmitters on 10 November 1994 and 15 November 1995 (*A* nest abandoned)

Penguin no.	Date	Eggs	Chicks	Adult mass (kg)	Chick mass (kg)
1	10 Nov 1994	2			
2			2		
1	22 Nov		1		
2			A		
1	26 Nov		A		
2			A		
17	15 Nov 1995		2	4.65	275/575
115			1	4.75	200
17	17 Nov		2	4.65	300/600
115			1	4.70	250
17	20 Nov		2	4.87	300/875
115	19 Nov		1	4.45	300

encountered on the nest. The nests were checked on 17 November, after a foraging trip of the equipped birds. The mass of the adults was almost unchanged, but the chicks in both nests had gained weight. We checked the nests one last time after the next foraging trip, on 20 November (nest 17) and 19 November (nest 115) and the chicks had again gained weight (Table 2). Transmitters operated until 29 December 1995 and 24 January 1996, respectively.

Satellite tracking

Locations obtained on Humboldt penguins 1 and 2 during the 1994/1995 breeding season (Fig. 1A and B, respectively), and on birds no. 17 and 115 during the 1995/1996 season (Fig. 1C and D, respectively) show no specific foraging areas. This is also true on a day-to-day basis (not shown). During both seasons, the penguins remained within 95 km of the coast (i.e. west of the island, Fig. 1D), and all foraging trips were within a strip extending from 31 km south (Fig. 1C) to 98 km north of the island (Fig. 1D). The maximum area covered by Humboldt penguins foraging from Pan de Azúcar Island during the breeding season is therefore 12 255 km^2.

During a typical foraging trip (Fig. 2, top), penguin 1 was located 16 times by Argos satellites. The bird left the island on the morning of 5 December, and was at sea when spotted at 0921 hrs, local time. It presumably remained at sea within 10 km of the coast until the evening (2005 hrs) and did not return to the colony, spending the night at sea. The next day the individual traveled southeast and then northeast until 2035 hrs, to spend a second night at sea in the same area as the night before. On the morning of 7 December the penguin swam west until 0754 hrs, then south (1018 hrs) and finally headed east to return to the island, where it arrived by 1841 hrs. During this 3-d foraging trip the penguin covered an area of 25 × 40 km, i.e. 1000 km^2.

Diving activity during the same trip was obtained via the salt water timer on the transmitter and is shown in Fig. 2 (bottom). According to this additional information, the individual left the island at ca. 0830 hrs on 5 December, i.e. just about 1 h before being spotted at sea by the satellite. Cumulative dive duration then increased over the course of the afternoon and reached 8.8 h (58%

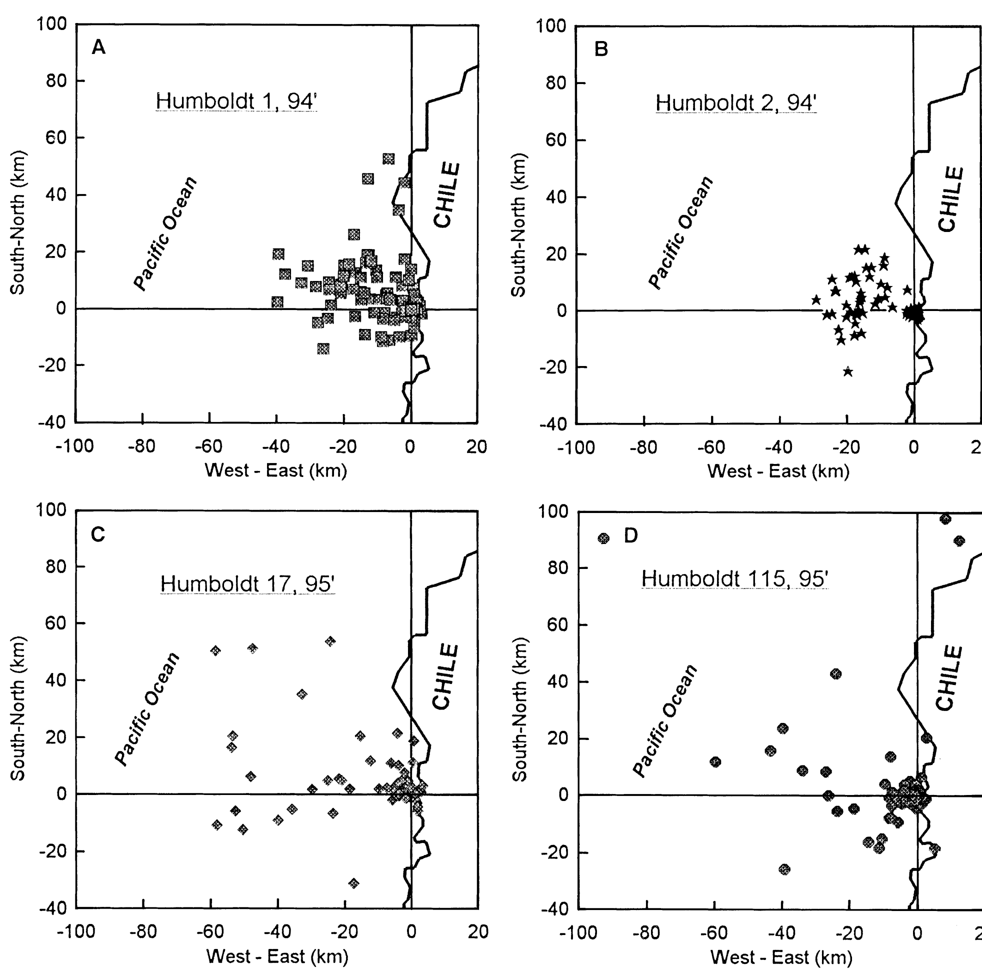

Fig. 1 *Spheniscus humboldti*. Distribution of all locations of Humboldt penguins during the 1994/1995 season (*top panels*) and the 1995/1996 season (*bottom panels*). Pan de Azúcar Island (26°S; 72°W) is located at the origin of the *x*- and *y*-axes

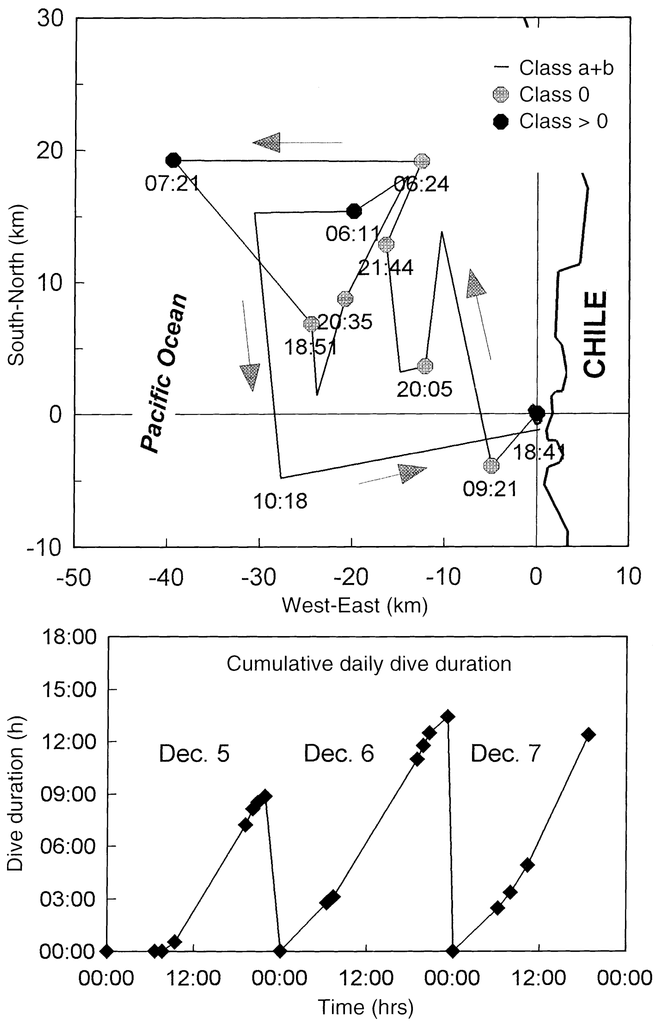

Fig. 2 *Spheniscus humboldti.* Details of a foraging trip of penguin no.1 for the period of 5 to 7 December 1994 (*top panel*). Location class indicated by symbol type. The bird covered an area of 1000 km^2. While the penguin was at sea, the salt water switch was timed and provided cumulative daily dive duration (*bottom panel*), which amounted to 13.5 h on 6 December. The platform transmitter terminal functioned continuously (whenever the bird was at the surface)

of time at sea) by midnight. The penguin therefore only spent ca. 6.5 h (42% of the time) at the surface between morning and midnight that day. On 6 December, the bird accumulated 13.5 h of underwater time (56% of time at sea). Cumulative daily dive duration follows a somewhat sigmoidal function during 6 December, indicating that Humboldt penguins dive less in the early morning and late at night. The individual again spent most of the early morning of 7 December at the surface. As on 6 December dive intensity increased after 1000 hrs, as shown by a steeper slope, and when returning to the island at 1841 hrs the bird had been underwater for 12.4 h (65% of time at sea) during that day.

A much less well resolved foraging trip is shown for penguin no. 2 for the period 1 to 3 January 1995 (Fig. 3, top). The transmitter on the bird was only on for two-thirds of the time, resulting in only seven locations during the entire trip. The penguin was located at sea at 0601 hrs and again at 2021 hrs on 1 January, 20 km west and 11.5 km north of the island. This bird also spent the night at sea and covered ca. 15 km between 2021 (1 January) and 0553 hrs on 2 January, when it was 26 km north and 17 km west of the island. It then returned toward the island and was located at the shore at 2138 hrs on 3 January. The total area covered by the bird during this foraging trip was 560 km^2, which may be an underestimate due to duty-cycling of the transmitter and resulting loss of foraging-trip resolution.

Cumulative dive durations (Fig. 3, bottom) again indicate that the penguin dived mostly in the afternoon before midnight. On 1 January the bird accumulated 13.4 h (75% of time at sea) of dive duration, most of it after 0900 hrs. Although the penguin spent the night at sea, it only dived occasionally, accumulating only 1.5 h of dive time by 0553 hrs. Diving activity (and the slope of the curve) increased thereafter, and by the evening

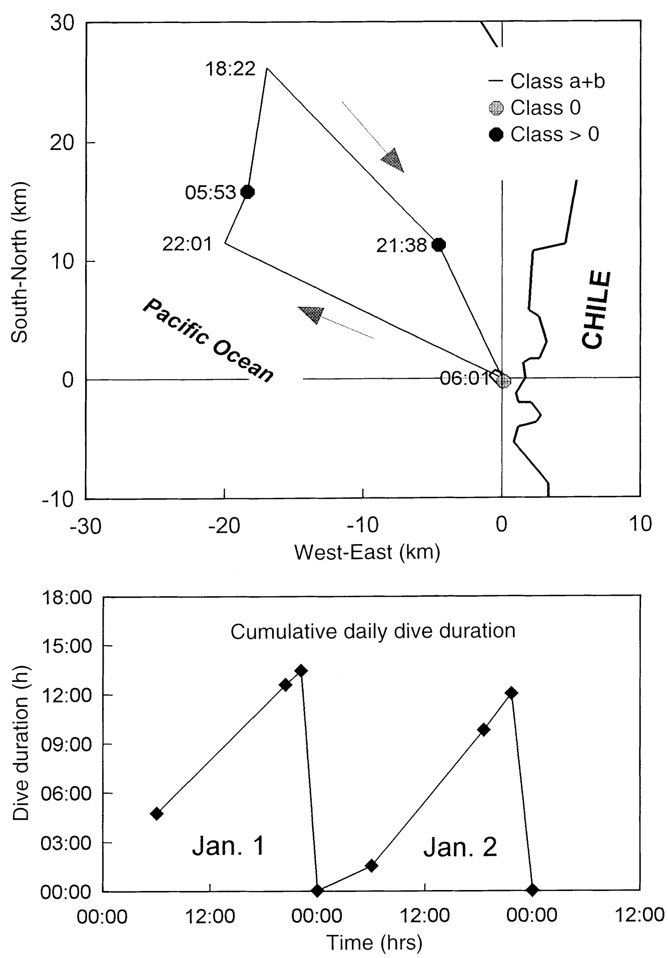

Fig. 3 *Spheniscus humboldti.* Foraging trip of bird no. 2 (1 to 3 January 1995) obtained with a programmed transmitter (16 h on, 8 h off). In comparison to Fig. 2, number of locations is lower, but time under water is comparable (13.5 h on 1 January). Location class is indicated by symbol type

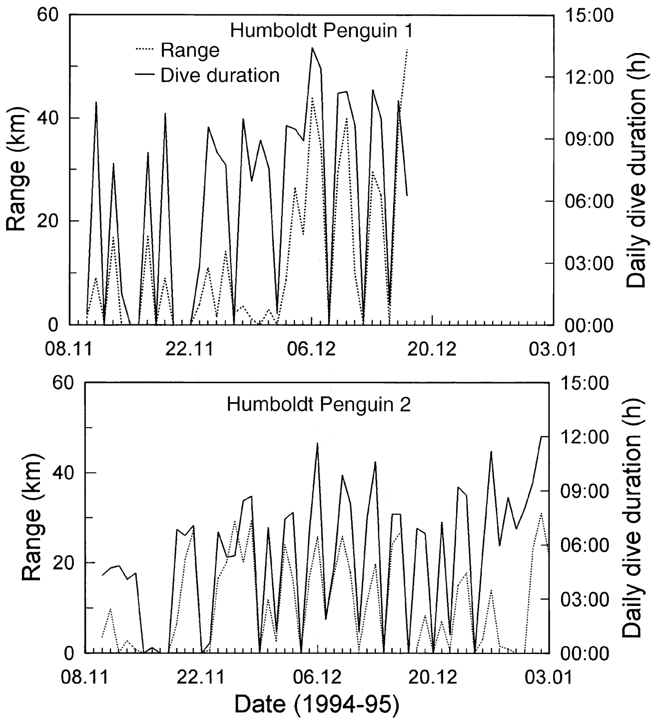

Fig. 4 *Spheniscus humboldti.* Daily dive duration (h) and foraging range (km) of both penguins were highly variable throughout the study period of 1994/1995, with variation caused by the duration of the interval between foraging trips (i.e. on land) and the different ranges chosen by both birds. There is no apparent trend

(2138 hrs) the penguin had accumulated 12 h (56% of time at sea) of dive time during that day.

Foraging activities throughout the breeding season

The data presented in Fig. 1 show the distribution of all the locations of the birds studied. However, the birds did not forage every day or within a similar range of the island. In order to detect a trend in daily activity, we plotted maximum range and daily dive duration versus date in Fig. 4. It becomes apparent, that penguin no. 1 (Fig. 4, top) only made foraging trips lasting 1 d before 22 November, interspaced with 1 to 2 d on land. Foraging was within a maximum range of 18 km from the island, and dive duration was $< 11\ h\ d^{-1}$ at sea. After 22 November, the duration of foraging trips was increased to 4 d, although the range still remained below 20 km until 2 December. That day, the bird left the island to forage as far as 40 km away, and spent 5 d at sea before returning. Daily dive duration was increased to 13 h during that trip. After 8 December, foraging range and trip duration decreased again, and the two subsequent trips were only 3 and 2 d long, respectively, within a range of 40 and 30 km, respectively.

Foraging trips of penguin 2 (Fig. 4, bottom) were of variable duration, from 4 d in early November to 1 d on 29 December, or 6 d between 4 and 11 December. The bird always remained within a maximal range of 30 km of the island. The duration of daily diving activity was between 5 h in early November and 12 h in January. Data of birds 17 and 115 (1995/1996 breeding season) could not be analysed similarly, due to the low number of locations obtained per day.

In order to detect a trend in foraging activity throughout the breeding seasons of 1994/1995 and 1995/1996, we computed the 4-d (maximum duration of foraging trips) running mean of the maximal range observed during each day (Fig. 5). While the range of bird no. 1 decreased from 7 to 2 km throughout November (Fig. 5, top), foraging range of bird no. 2 increased from 1 to 26 km during the same time. This inverse correlation between the foraging ranges of the two birds was also maintained after 3 December, when mean daily foraging range of bird no. 1 increased to 35 km, while that of bird no. 2 was within 17 km of the island. Mean daily foraging ranges calculated for penguins 17 and 115 during the breeding season of 1995/1996 (Fig. 5, bottom) also lack similarity, and there was no conspicuous trend shown by both birds throughout the period of investigation. Bird no. 115 reached maximum mean distances from the island on 6 and 19 January 1996, with 84 and 92 km, respectively.

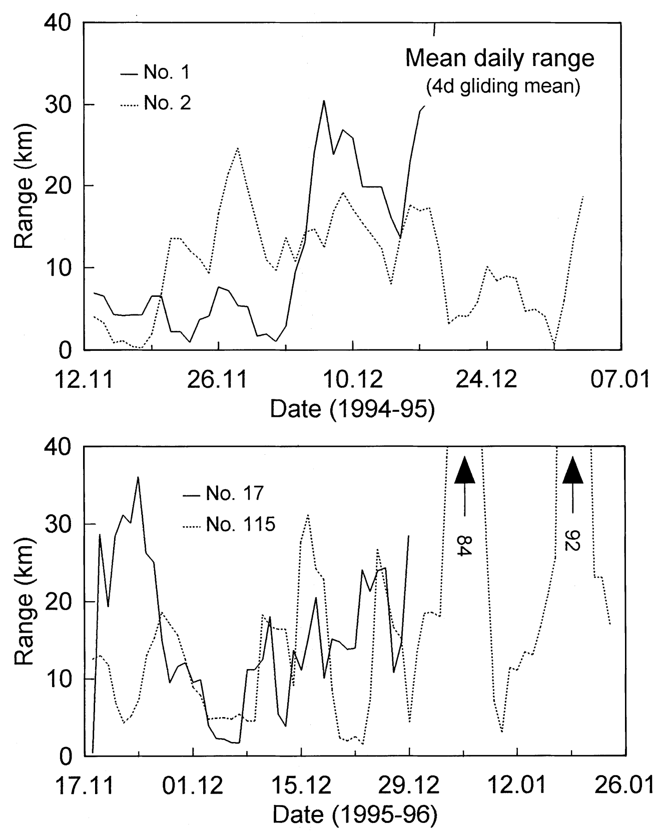

Fig. 5 *Spheniscus humboldti.* When smoothing daily range over a 4-d period, penguins no. 1 and 2 seem to forage anticyclical to each other in 1994/1995 (*top panel*), while there is no trend for birds 17 and 115 during the 1995/1996 season (*bottom panel*)

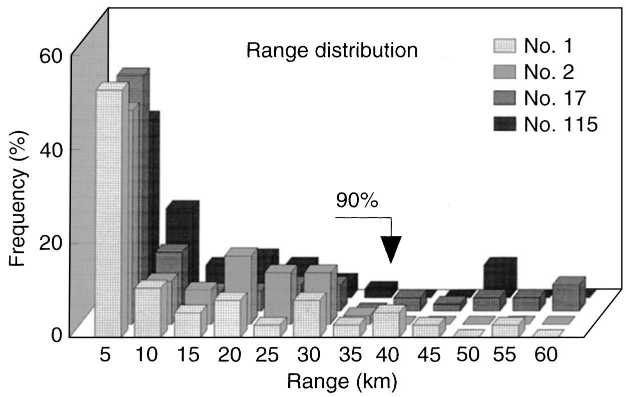

Fig. 6 *Spheniscus humboldti*. Penguin distribution at sea was computed from all locations. Of all birds 50% were observed within 5 km of the island, while the 90% limit was 35 km from the island

The distribution of the maximum ranges of the four penguins during both breeding seasons was remarkably similar (Fig. 6). Most of the locations occurred in the vicinity of the island, because this was also where the penguins went to rest, thus presumably increasing the likelihood of successful data transmission to the satellites. In 50% of the time the locations obtained ($n = 512$) were within a radius of 5 km of the centre of Pan de Azúcar Island. Of all locations 90% were within 35 km and 95% were within 50 km of the island.

Mean travelling speed during foraging trips

Mean travelling speed of the penguins was determined from location data (class 0 and better) by dividing the distance swum between two locations by the elapsed time. This is not the same as swimming speed, since time between two consecutive locations was too long to resolve in detail the route taken by the birds. Furthermore,

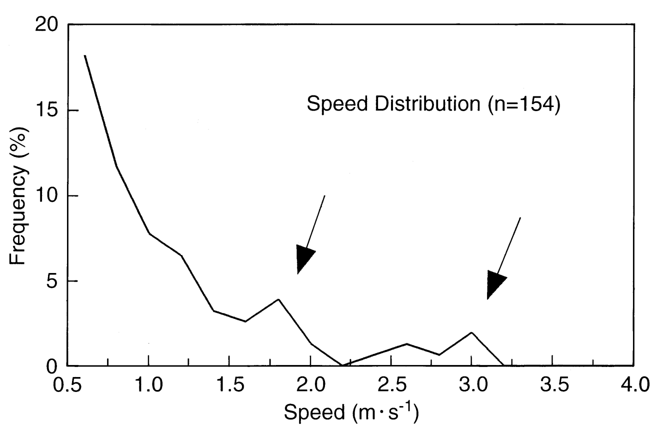

Fig. 7 *Spheniscus humboldti*. Although travelling speed computed from satellite data is a very coarse estimate of swimming speed, frequency distribution of all computed speeds showed peaks at 1.8 m s^{-1} (presumably travelling speed) and 3 m s^{-1} (presumably porpoising, see "Discussion – Mean travelling speed during foraging trips")

computation of average speed of movement does not include underwater swimming, which may have followed a more complicated route, i.e. during feeding. Consequently, mean travelling speed computed from the pooled data (all birds, but excluding nights and birds on the island when travelling speeds were <0.2 m s^{-1}) is 0.94 m s^{-1} (SD 1.14, $n = 154$). A plot of travelling speed frequency distribution (Fig. 7) shows two peaks, indicated by arrows, at 1.8 and at 3 m s^{-1}.

Discussion and conclusions

Device effects

Satellite transmitters used in the present study were smaller (volume 134 cm^3, cross-sectional area only 3.5% of bird cross-sectional area), and lighter (2.8% of bird body mass) than those used by Davis and Miller (1992), and they were streamlined according to the suggestions in Bannasch et al. (1994). However, although we minimized device-induced effects according to Wilson and Culik (1992), two of the birds equipped in 1994 abandoned their nests within 12 and 16 d of attachment, respectively. While Humboldt penguins are known to "move" from one burrow to another during the breeding season (authors' personal observations), we were not able to locate the new nest sites (if any). If physical impairment by the PTTs was not the main cause for the observed nest desertions, the reason for these may have been human disturbance (i.e. Culik and Wilson 1995). During the period of our 1994 investigations, the island was visited repeatedly by up to six people filming the wildlife for a TV production. No visitors were observed in 1995, when equipped penguins did not abandon their nests during the 5 d this was monitored, although in addition to wearing the transmitters, birds were weighed and blood samples taken twice at 2-d intervals after PTT attachment. We will attempt to show in the section on foraging effort that another reason for the difference between birds studied in 1994 and 1995 may have been food availability.

Performance of satellite transmitters

The maximum number of locations obtained from bird no. 1 was 7 d^{-1}, which corresponds to the number of satellite passes at that latitude. The lower daily mean (4.2 locations d^{-1}) may be attributable to diving activity at times when the satellite was overhead. While Argos requires the reception of at least two perfect messages (whole antenna "visible") in order to compute locational information, this may be difficult to achieve during the 10 min period the satellite "sees" the area. Considering that transmitter no. 2 was only "on" two-thirds of the time, the number of daily locations expected from that PTT, when compared to no. 1 would be 2.8 d^{-1}. Because the number of locations received was 3.1 or about 10%

higher, we assume that this can be explained by a larger proportion of the "on" period being shifted into resting phases of the bird, thus coinciding with good transmission periods.

The reason for the low number of daily locations obtained from birds no. 17 and 115, on the other hand, could have been the unfavourable starting time of the transmitters, with a larger proportion of the "on" period being shifted into the active at-sea phase of the birds. The reasoning behind this is as follows: if penguin location by satellite is successful during every overpass when the birds are not actively foraging, but rather pausing at the surface, and if such periods of rest occur more frequently at night, i.e. between 2000 and 0400 hrs local time, then the reduction in the mean number of locations obtained from bird no. 1 (continuously operating transmitter, but no. of locations reduced to 60% as opposed to the maximum of 7 d^{-1}) would only have occurred during the day, between 0400 and 2000 hrs. In other words, bird activities during the day would have reduced the number of successful locations to 40% of the maximum, or 1.86 locations in 16 h. From penguins no. 17 and 115 (PTTs started at 0400 hrs local time) we actually obtained 1.86 and 1.68 positions d^{-1}, respectively, which compares well. This is evidence that (1) Humboldt penguins spend significantly more time at the surface or on land during the night and (2) the success of satellite tracking in marine animals is directly linked to the time-activity budgets of these animals.

Mean travelling speed during foraging trips

Travelling speeds (averaging swimming speed, surface pauses and meandering tracks) measured from few daily satellite locations only give a very simplified overall average. The reason for this is the low resolution of the foraging trips. Consequently, we would expect that our estimate of travelling speed (0.94 m s^{-1}) is lower than estimates based on other, more detailed information. Using radio telemetry, Trivelpiece et al. (1986) found the travelling speed of similar-sized Gentoo penguins (*Pygoscelis papua*) to be 1.2 m s^{-1}, and a comparable study on African penguins (*Spheniscus demersus*) also yielded a mean travelling speed of 1.2 m s^{-1} (Heath and Randall 1989). However, Davis and Miller (1992) found again 1.2 m s^{-1} in Adélie penguins, although they used satellite telemetry. This may be due to the fact that at their study latitude, Argos provides up to 28 locations d^{-1}, which compares to the resolution obtained via radio telemetry.

Travelling speed distribution of Humboldt penguins (Fig. 7) shows peaks at 1.8 and 3 m s^{-1}, which may relate to the preferred swimming speeds of these birds. While the first peak (1.8 m s^{-1}) is within the normal underwater speed range of Humboldt penguins (1.55 to 2.2 m s^{-1}; Wilson and Wilson 1990), the second peak (3 m s^{-1}) rather corresponds to speeds observed during porpoising (2.8 to 4 m s^{-1} in African penguins; Wilson and Wilson 1990) via visual observation or radio tracking. From this we conclude that satellite telemetry can provide useful information on animal swimming speed, if sufficient data can be obtained.

Daily foraging activity

From the number of daily locations obtained via Argos satellites, we estimated (see above) that the main period of Humboldt penguin foraging activity was during the daylight hours. This is further supported by the analysis of cumulative daily dive duration (Figs. 2, 3). The sigmoidal curve shape indicates that Humboldt penguins dive less at night than during the day. These conclusions agree with the findings of Wilson and Wilson (1990) and more recently with those of Luna-Jorquera et al. (1997b) who reported that Humboldt penguins have a diurnal pattern of foraging behaviour, going to sea at dawn, spending the day at sea and returning to the colony at dusk or remaining at sea for another day. Similar observations were made on non-nesting African penguins by Frost et al. (1976). Although it might be expected that penguins can obtain more food by foraging at night, due to the rise of the scattering layer, it has been demonstrated by Wilson et al. (1993) that several penguin species are visual hunters, requiring light for prey perception.

Daily dive duration computed from our data cannot be compared with foraging trip duration obtained by other researchers on species of *Spheniscus* penguins using VHF-radio tags (e.g. Heath and Randall 1989). The reason is that satellite tracking does not yield the exact times of departure and return of the birds from the breeding island. Wilson et al. (1989), however, using speed meters accumulating total dive time on photographic film found mean dive durations of 10 h d^{-1} during the breeding season in African penguins and 7.1 h d^{-1} in Humboldt penguins. The data we obtained from birds no. 1 and 2 compares well (data from 17 and 115 could not be analysed because of too few locations to subdivide dive duration on a day-to-day basis, but see below). We found that bird no. 1 spent on average 9.0 h submerged per day (range 1.5 to 13.5 h, 25 d at sea) while bird no. 2 spent on average 7.8 h submerged per day at sea (range 2 to 13.5 h, 36 d at sea). Wilson and Wilson (1990) reported that in African penguins, dives of 105 s duration are followed by mean surface times of 19 s, i.e. that foraging activity includes 18% of surface time. This would entail, that Humboldt penguins diving for a maximum of 13.5 h daily spend at least 16 h actively foraging, if surface time between dives is included. This is the total duration of daylight hours at 26°S, in the summer.

Foraging range

Foraging ranges of penguins during the breeding season reflect the availability and proximity of their major food sources and the constraints imposed by the needs of their chicks. Consequently, foraging ranges can be highly variable within the same species (e.g. Kerry et al. 1995). The foraging ranges of Humboldt penguins from Pan de Azúcar Island were variable both in time and between birds (Fig. 5) and ranged between 2 and 92 km (4-d means). The only other figure for Humboldt penguins stems from Wilson et al. (1989). They found that these birds travel on average 32.2 km when foraging. This, however, cannot easily be converted into range, as travelling distance includes foraging dives (i.e. vertical distance) as well as changes in heading associated with prey localisation. Humboldt penguins from Pan de Azúcar Island did not seem to prefer particular areas, but their distribution at sea allowed us to determine the distance from the island where they were most likely to be encountered (Fig. 6). In total, 90% of all satellite locations came from a range of 35 km around the island (50% within 5 km). This compares well to estimates from African penguins which were found to forage within 15 to 40 km of their island (Heath and Randall 1989; Wilson et al. 1989) and to results of Wilson et al. (1988), who counted African penguins at sea during the breeding season and found 50% of the birds within 3 km of the coast.

The close proximity of Humboldt penguin foraging areas to the Chilean coast is related to the distribution of their main prey, clupeid-type pelagic school fish such as the anchoveta (*Engraulis ringens*). This pelagic school fish occurs in variable school sizes, densities being highest in the vicinity of the Pacific coast of South America, due to upwelling of the cold and nutrient-rich Humboldt current (Duffy 1989). Consequently, fishery activities are also concentrated along the coast, in the same areas used by foraging Humboldt penguins. The Chilean industrial fishery fleet comprises 578 vessels, added to which 10 864 vessels are operated by "traditional" fishermen (SERNAP 1994). In total, 2.7 million metric tons of anchoveta were caught in 1994, which amounts to 25% of total Chilean landings during that year. While the majority of the anchoveta were caught in northern Chile, ca. 25% of the annual catch came from the regions adjacent to Pan de Azúcar Island (SERNAP 1994). From this, the questions that need to be addressed in the future are: do fisheries affect Humboldt penguin behaviour at sea, is at-sea behaviour affected by prey availability or other factors, and how can this be measured?

Foraging effort and sea surface temperature anomaly

Prey encounter rate of penguins depends on the volume of water searched per unit time (Wilson and Wilson 1990). Therefore, assuming that penguins need to catch a minimum of food within a certain time prior to returning to the colony, the birds may vary swimming speed and diving depth to increase searching efficiency (Wilson and Wilson 1990). However, when food availability is low, penguins might be increasing foraging effort, by increasing their foraging range and daily dive duration. In this study, the low number of daily localisations did not enable us to determine daily swimming speed, and dive depth was not measured. However, if birds increased foraging effort throughout the study period, this should have been measurable.

As shown by Fig. 5, mean daily foraging range (4-d mean) of Humboldt penguins showed no apparent trend during the 1994/1995 or 1995/1996 seasons and was not correlated between the birds. Wilson and Wilson (1990) report that *Spheniscus* spp. penguins are sociable at sea and often form groups. However, they seem to split up when foraging, as "foraging *Spheniscus* penguins are most frequently encountered as singletons" (Wilson and Wilson 1990). Consequently, birds explore individual patches of prey, which is reflected by the number of hours spent diving per day (Fig. 4) which does not correlate between birds. If Humboldt penguins showed no trend in foraging range nor daily dive duration, what else could have been increased?

Penguins can spend proportionally more time at sea, by reducing the duration of periods spent on land between foraging trips. In order to test whether penguins from Pan de Azúcar varied time at sea during both study periods in relation to prey availability, we computed the 4-d mean of daily dive duration for each bird. This parameter was then correlated with an independent measurement of marine productivity, taken to be the so-called sea surface temperature anomaly (SSTA, in °C) in the region. SSTA data were provided by the Servicio Hydrográfico y Oceanográfico de la Armada de Chile (SHOA, Valparaíso, Chile) for the harbour of Caldera, 100 km south of Pan de Azúcar Island. In brief, SSTA data are computed by comparing the mean monthly sea surface temperatures to the previous 10-year average for that month. It is expected, that SSTA data together with sea level reflect oceanographic conditions related to the Humboldt Current, i.e. thermocline and nutrient availability (Gates et al. 1990). Penguin data were also related to fishery landings at Caldera (obtained from Servicio Nacional de Pesca, Caldera).

During the 1994/1995 season, SSTA showed a strong positive anomaly all along the Pacific coast of South America (CPPS 1996). In December, SSTA values reported for the Niño 1 to 4 regions and the coastal zone averaged 1 °C higher than normal. Similarly, sea level measurements at Caldera were 40 cm above average (CPPS 1996). SSTA and fishery landings were negatively correlated (Pearson correlation analysis: $R = -0.998$, $p < 0.001$), with large fluctuations in both parameters throughout the study period (Fig. 8, top). Mean Humboldt penguin foraging effort (4-d mean daily dive duration) was highly correlated ($R = 0.675$, $p < 0.001$, $n = 37$) between the two birds (no. 1 and 2, Fig. 8, bottom). This seems to indicate that both birds reacted simultaneously to some extrinsic factor. Correlation of foraging effort between penguins and fishery landings was negative ($R = -0.79$ and -0.68 for birds no. 1 and 2, respectively, $p < 0.001$), indicating that birds increased

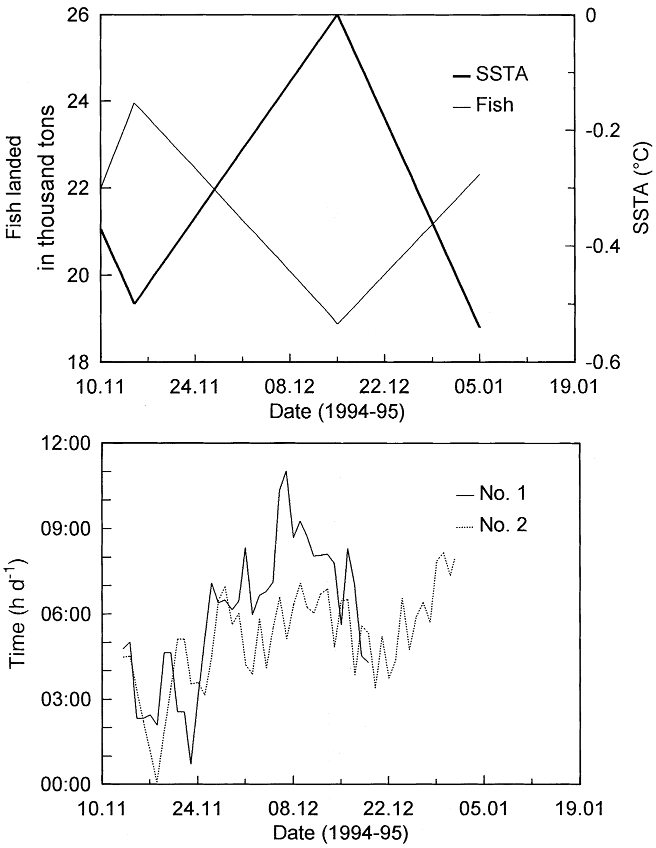

Fig. 8 *Spheniscus humboldti*. Sea surface temperature anomaly (*SSTA*, °C) and fishery landings at Caldera, Chile, are negatively correlated for the 1994/1995 season (*top panel*). The clear trend shown by the mean daily dive duration (*bottom panel*) is correlated with SSTA and negatively correlated with fishery landings

foraging effort at a time when fish availability presumably decreased. Finally, foraging effort was positively correlated to SSTA values ($R = 0.79$ and 0.69, $p < 0.001$ penguins 1 and 2, respectively), and this suggests that increasing sea surface temperatures may influence prey availability at a time of high sea levels. These adverse conditions may also have been the cause for the observed nest desertions.

As opposed to this, the breeding season of 1995/1996 was characterized by SSTA conditions moderately (0.5 °C) colder than normal, and SSTA and sea level signals were weak (CPPS 1996), fluctuating only between −0.8 and −1.2 °C and 130 and 120 cm, respectively. Fishery landings only partly paralleled the SSTA trend, but correlation was significant (Pearson correlation: $R = -0.73$, $p < 0.001$, $n = 52$). However, foraging efforts (i.e. mean daily dive duration) of birds no. 17 and 115 were not significantly correlated with each other ($R = 0.37$, $p = 0.09$) and did not correlate with fishery landings. While daily dive durations of bird no. 17 were inversely correlated to SSTA ($R = -0.43, p = 0.03$), those of bird no. 115 were not ($R = 0.061$).

From these comparisons, it would appear that SSTA could be used as a predictor of penguin at-sea activity, if the signal during a study period is sufficiently strong, i.e. >0.4 °C, if the anomaly is positive and if it correlates with an increase in sea level. Mean daily dive time appears to be a useful measure of overall foraging effort. It seems that when prey availability is low, Humboldt penguins maximize time at sea rather than search radius. Further investigations using time/depth recorders should show whether the birds also increase foraging efficiency (i.e. swimming speed and dive depth) under these conditions.

Conclusions

Our study provides evidence to support the hypothesis that even small SSTA increases are associated with deteriorating conditions of prey or fish availability, and yield an increase in foraging effort of Humboldt penguins. Because penguins forage mainly during the daylight hours, and because daily time under water reaches values as high as 13.5 h, we assume that under these circumstances the birds are working with maximal foraging effort. Humboldt penguins foraging from Pan de Azúcar Island showed no preference for particular marine sectors and 90% of the birds remained within 35 km of their breeding sites. From this result, we suggest that the park authorities delimit a 35 km protection zone around the island, which corresponds to the entire coastal area of Pan de Azúcar National Park. If no measures for the protection of the Humboldt penguin are adopted, we fear that a future El Niño will have a catastrophic impact on a species, which, since the last event, has not been allowed to build up population reserves.

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